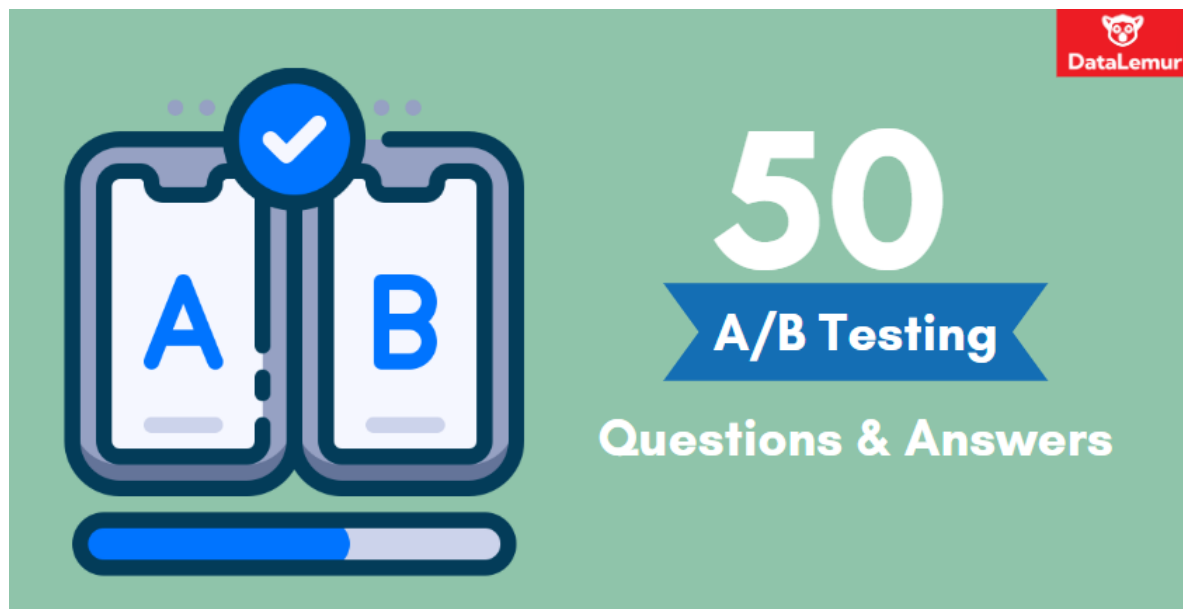


# 50 A/B Testing Interview Questions & Answers

A/B testing interview questions appear in about ~50% of data science interviews, especially for [Product Data Science roles](#) at consumer-tech companies like [Meta](#), Airbnb, and Uber. To help you prepare, we've curated a list of 50 A/B Testing Interview Questions and Answers broken into the five main types of questions:

- Experimental Design Questions
- Metric Selection Questions
- Interpretation of A/B Test Results Questions
- Statistical Power Calculation Questions
- Multiple Testing Questions



## Section 1: Experimental Design Interview Questions

**1. How do you determine the duration of an A/B test?** To determine the duration of an A/B test, consider the following factors:

- Sample size and statistical significance: The primary factor in determining test duration is reaching a statistically significant result. You need a large enough sample size in each variation to confidently conclude that the observed differences are not due to chance.
- Business cycle and seasonality: Consider your business cycle and seasonality when determining test duration. For example, if you're an

e-commerce site like [Amazon](#), you may need to run tests for at least a full week to capture behavior across weekdays and weekends.

- User behavior and purchasing cycle: Think about your typical user behavior and purchasing cycle. If you're testing a change related to a high-consideration purchase with a long decision cycle, you may need to run the test for several weeks to fully capture the impact on conversions.
- Minimum detectable effect: The smaller the minimum improvement you want to be able to detect, the larger the sample size needed and thus the longer the test duration. If you only care about detecting large effects, you can reach significance faster.

## 2. What are some common pitfalls to avoid when designing an A/B test? Common pitfalls in A/B test design include:

- inadequate sample sizes
- biased sampling methods
- insufficient randomization
- running too many experiments at once

In an interview, you usually want to contextualize your answer about A/B testing pitfalls to the business & team at-hand. For example, if you were [interviewing at Uber](#) on the Driver Growth division, here are some specific A/B testing issues you might encounter:

- Difficulty isolating variables: Driver behavior is influenced by many external factors like local market conditions, seasonality, competitor activity, etc. This can make it challenging to isolate the impact of a specific A/B test variable.
- Long time to reach statistical significance: Given the long-term nature of driver acquisition and retention, it may take months for a test to reach statistically-significant results on metrics like driver retention and lifetime value
- Potential interference between simultaneous tests: With multiple teams likely running A/B tests concurrently on different aspects of the driver experience (e.g. signup flow, incentives, app features), there is risk of tests interfering with each other and confounding results.
- Ethical considerations with underserved segments: If an A/B test inadvertently provides a worse experience to certain underserved driver segments, even if unintentional, it could have outsized negative

driver segments, even if unintentional, it could have outsized negative impact on those groups.

## Uber Driver Growth A/B Testing Pitfalls

- **Potential interference between simultaneous tests**
- **Ethical considerations with underserved segments**
- **Long time to reach statistical significance**
- **Difficulty isolating variables**



**3. How would you ensure randomization in an A/B test?** Randomization in an A/B test can be ensured by randomly assigning participants to treatment and control groups, thereby minimizing the risk of bias and ensuring that the groups are comparable.

**4. Can you explain the concept of bucketing in the context of A/B testing?** Bucketing refers to the process of assigning participants to treatment and control groups based on predetermined criteria, such as geographic location, device type, or user segment.

**5. What considerations should be made when selecting the sample size for an A/B test?** Sample size for an A/B test should be determined based on considerations such as the desired level of statistical power, expected effect size, baseline conversion rate, and significance level.

**6. What is a control group, and why is it important in A/B testing?** The control group serves as a baseline for comparison, allowing researchers to assess the impact of the treatment by comparing outcomes between the treatment and control groups.

**7. How would you handle variations in user behavior over time during an A/B test?** Variations in user behavior over time can be addressed by conducting the test over a sufficient duration, ensuring that the test period covers different days of the week, times of day, and user segments.

**8. Describe the process of creating treatment groups for an A/B test.**

Treatment groups can be created by randomly assigning participants to different experimental conditions or by using stratified sampling methods to ensure that each group is representative of the population. Usually the in-house A/B testing framework at a company like Facebook or Uber is able to do this for you, automatically!

**9. What measures can be taken to minimize the impact of external factors on the results of an A/B test?** External factors can be minimized by conducting the test in a controlled environment, implementing safeguards to prevent interference, and monitoring external events that may impact the results.

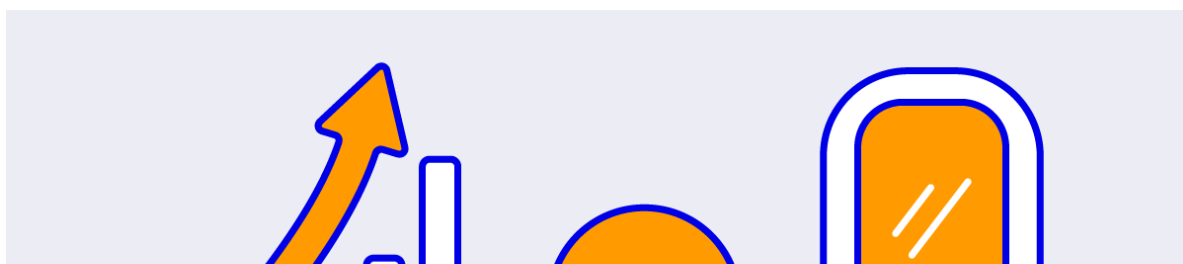
**10. How would you determine the statistical significance level for an A/B test?** The statistical significance level, often denoted as alpha ( $\alpha$ ), is typically set at 0.05 or 0.01, indicating the acceptable probability of falsely rejecting the null hypothesis.

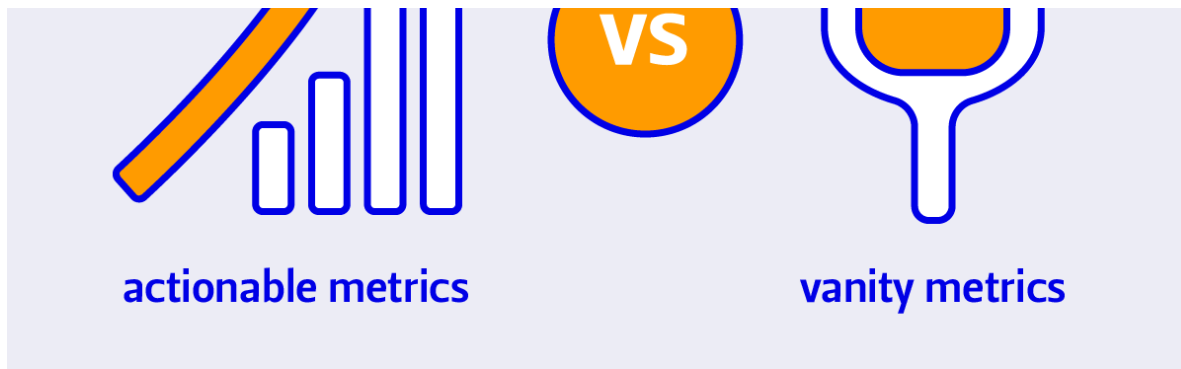
## **Section 2: Metric Selection Interview Questions**

**11. What criteria would you use to choose appropriate metrics for an A/B test?** Appropriate metrics for an A/B test should be relevant to the business objectives, sensitive to changes in the treatment, reliable, and actionable.

**12. Can you differentiate between primary and secondary metrics in A/B testing?** Primary metrics are directly related to the primary goal of the experiment, while secondary metrics provide additional insights or context but are not the primary focus.

**13. How would you prioritize metrics when they conflict with each other in an A/B test?** Prioritization of metrics should consider their alignment with the primary goals, sensitivity to changes, reliability, and practical relevance to the business.





**14. What are vanity metrics, and why should they be avoided in A/B testing?** Vanity metrics are superficial metrics that may be misleading or irrelevant to the business objectives and should be avoided in A/B testing.

For example, imagine you were interviewing for a Product Data Science role at Meta, and had a question about key metrics to track for Facebook Groups. Here's some potential vanity metrics to avoid mentioning to your interviewer:

- Total number of Groups: Tracking the total number of Groups on the platform might seem important, but it doesn't necessarily reflect the health or engagement of those Groups. Many could be inactive or low-quality.
- Total number of Group members: Similar to total number of Groups, tracking total Group membership doesn't account for member activity or engagement. A Group could have many members but low participation. Focusing on this could lead to tactics that drive superficial member growth without improving the Group experience.
- Number of Group posts: Measuring the raw number of posts in Groups doesn't consider the quality, relevance, or value of those posts. This metric could be gamed by encouraging low-effort, spammy posting just to drive up the numbers, rather than facilitating meaningful conversations.

**15. How do you ensure that the selected metrics are relevant to the business goals?** Selected metrics should directly reflect the impact of the treatment on the desired outcomes, such as conversion rate, retention rate, revenue, or user satisfaction.

**16. Explain the difference between leading and lagging indicators in the context of A/B testing.** Leading indicators are predictive metrics that signal future outcomes, while lagging indicators are retrospective metrics that reflect past performance

For example, imagine you were interviewing to be a Data Scientist on Airbnb's Pricing Team. Some leading indicators you could bring up:

- Number of hosts viewing the new pricing recommendations: This measures initial engagement with the new pricing feature and predicts future adoption.
- Percentage of hosts accepting the pricing suggestions: This indicates the perceived relevance and trustworthiness of the recommendations, predicting future usage.
- Change in average listing price: This immediate shift can predict the eventual impact on bookings and revenue.

Lagging Indicators to bring up for the Airbnb Data Scientist Interview:

- Host retention and lifetime value: The long-term impact on host satisfaction and retention on the platform is crucial, but will significantly lag the initial pricing changes.
- Guest reviews mentioning price: An eventual lagging indicator of guest price perception and satisfaction, which could impact rebookings and word of mouth.

**17. How would you handle situations where the chosen metrics may be influenced by external factors?** External factors influencing the metrics should be identified and controlled for, or alternative metrics should be selected that are less susceptible to external influences.

**18. What role does statistical power play in metric selection for A/B testing?** Statistical power considerations should be taken into account when selecting metrics to ensure that they are sensitive enough to detect meaningful differences.

**19. Can you provide examples of quantitative and qualitative metrics used in A/B testing?** Examples of quantitative metrics include conversion rate, revenue per user, and average session duration, while qualitative metrics include user satisfaction ratings and feedback.

**20. How would you measure user engagement in an A/B test?** User engagement can be measured using metrics such as session duration, number of page views, click-through rate, or interaction frequency.

## **Section 3: Interpretation of A/B Test Results**

### **Interview Questions**

**21. What steps would you take to validate the results of an A/B test**

Validation of A/B test results involves cross-checking with other data sources, conducting sensitivity analyses, and ensuring that the observed effects are consistent and robust.

**22. How do you differentiate between statistically significant results and practical significance in A/B testing?**

Statistical significance alone does not guarantee practical significance; it is essential to consider the magnitude of the effect and its potential impact on the business objectives.

**23. What factors could lead to false positives or false negatives in the results of an A/B test?**

False positives may occur due to random chance or multiple testing, while false negatives may result from inadequate sample sizes or insufficient statistical power.

**24. Can you explain the concept of effect size and its relevance in interpreting A/B test results?**

Effect size quantifies the magnitude of the difference between treatment groups and provides context for interpreting the practical significance of the results.

**25. How would you communicate the findings of an A/B test to stakeholders?**

Communication of A/B test findings should be clear, concise, and tailored to the audience, highlighting key insights, implications, and next steps.

**26. What considerations should be made when comparing the performance of multiple variants in an A/B test?**

Comparison of multiple variants should consider both statistical significance and practical significance, as well as potential trade-offs between different performance metrics.

**27. How do you assess the robustness of A/B test results against variations in data distribution?**

The robustness of A/B test results can be assessed by conducting sensitivity analyses, testing alternative hypotheses, and examining the consistency of results across subgroups.

**28. What role does confidence interval play in interpreting the uncertainty of A/B test results?**

Confidence intervals provide a range of plausible values for the true effect size, accounting for uncertainty in the estimate.

**29. How would you handle situations where the results of an A/B test are inconclusive?** Inconclusive results may occur due to insufficient sample sizes, unexpected variations in user behavior, or limitations in the experimental design.

**30. Can you discuss the importance of considering practical constraints and ethical implications in interpreting A/B test results?** Consideration of practical constraints and ethical implications is crucial for interpreting A/B test results responsibly and making informed decisions.

## **Section 4: Statistical Power Calculation Interview Questions**

**31. What factors influence the statistical power of an A/B test?**

Factors influencing the statistical power include sample size, effect size, significance level, and variability in the data.

**32. How would you calculate the statistical power for a given A/B test scenario?** Statistical power can be calculated using statistical software or online calculators based on the desired level of significance, effect size, and sample size.

**33. Can you explain the relationship between sample size, effect size, and statistical power?** Sample size, effect size, and statistical power are interrelated, with larger sample sizes and effect sizes leading to higher statistical power.

**34. How does the significance level affect the statistical power of an A/B test?** The significance level, typically set at 0.05 or 0.01, determines the threshold for rejecting the null hypothesis and affects the statistical power.

**35. What measures can be taken to increase the statistical power of an A/B test?** Increasing the sample size, choosing more sensitive metrics, or reducing variability in the data can help increase the statistical power of an A/B test.

**36. Can you discuss the trade-offs between statistical power and Type I error rate in A/B testing?** Trade-offs between statistical power and Type I error rate involve balancing the risk of false positives with the risk of false negatives.



**37. How would you determine the appropriate effect size for**

**calculating the statistical power?** The appropriate effect size for calculating statistical power depends on the context of the experiment and the magnitude of the expected difference between groups.

**38. What role does variability in the data play in estimating the**

**statistical power?** Variability in the data, measured by standard deviation or variance, influences the precision of estimates and, consequently, the statistical power.

**39. Can you provide examples of scenarios where a low statistical**

**power could lead to misleading conclusions?** Low statistical power increases the risk of Type II errors, where true effects may go undetected due to insufficient sample sizes.

**40. How do you interpret the results of a power analysis in the**

**context of A/B testing?** Interpretation of power analysis results involves assessing whether the chosen sample size provides adequate sensitivity to detect meaningful differences with a desired level of confidence.

For more on Power Calculations read [this publication](#) by the Boston University School of Public Health.

## **Section 5: Handling Multiple Testing Interview Questions**

**41. What is multiple testing, and why is it a concern in A/B testing?**

Multiple testing refers to the practice of conducting multiple statistical comparisons simultaneously, leading to an increased risk of false positives..

**42. How do you control the family-wise error rate in multiple testing**

**scenarios?** Family-wise error rate control methods, such as Bonferroni correction or Holm-Bonferroni method, adjust the significance threshold to account for multiple comparisons.

*The original p value*



$$\text{Bonferroni-corrected } p \text{ value} = \frac{\alpha}{n}$$

*The number of tests performed* 

**43. Can you explain the Bonferroni correction and its application in A/B testing?** The Bonferroni correction divides the significance level by the number of comparisons to maintain the overall Type I error rate at the desired level.

**44. What are some alternative methods for controlling the Type I error rate in multiple testing?** Alternative methods for controlling Type I error rate include false discovery rate (FDR) control and sequential testing procedures.

**45. How would you adjust the p-values for multiple comparisons in an A/B test?** P-values can be adjusted using methods such as the Benjamini-Hochberg procedure or the Šidák correction to account for multiple comparisons.

**46. Can you discuss the trade-offs between different approaches to multiple testing correction?** Trade-offs in multiple testing correction involve balancing the risk of false positives with the potential loss of statistical power due to stringent correction methods.

**47. What considerations should be made when interpreting results after multiple testing corrections?** Interpretation of results after multiple testing corrections should consider both statistical significance and practical significance, as well as potential biases or confounding factors.

**48. How do you determine the appropriate correction method based on the specific A/B test scenario?** The appropriate correction method depends on factors such as the number of comparisons, the correlation structure of the data, and the desired balance between Type I and Type II error rates.

**49. Can you provide examples of situations where failing to correct**

**for multiple testing could lead to erroneous conclusions?** Failure to correct for multiple testing can lead to an inflated Type I error rate and erroneous conclusions about the significance of the results.

**50. How do you communicate the implications of multiple testing corrections to stakeholders?** Communication of the implications of multiple testing corrections to stakeholders involves explaining the rationale behind the correction methods and the impact on the interpretation of the results.

## More Statistics Interview Questions

Besides A/B Testing questions, to prepare for the Data Science interview test yourself on some probability & statistics questions in this article on the [Top 20 Statistics Questions asked in the Data Science Interview](#).

A/B testing shows up often in Data Science interviews, especially during Product-Sense Interview Rounds. Learn all about it in our [Product-Sense Interview Guide](#).

You can also practice a few of these stats & product-sense data questions interactively, alongside the DataLemur community:

1. [Frequentist vs. Bayesian](#)
2. [Surprising Decrease](#)

## How to Prepare for the Data Science Interview?

It's not just statistics and A/B Testing sections that are a must to prepare: data interviews cover a TON! Test yourself and solve over 200+ [data interview questions on Data Lemur](#) which come from companies like Facebook, Google, and VC-backed startups.

 Nick Singh 

### SQL & Data Science Interview Questions

Practice the most common SQL, Statistics, ML, and Python questions asked in FAANG Data Science & Data Analyst interviews!

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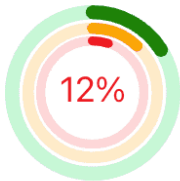
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I'm a bit biased, but I also recommend the book [Ace the Data Science Interview](#) because it has multiple FAANG A/B testing, statistics, product-sense, and case study interview questions!